

Augment, Replace, or Simulate? A Systematic Review and Empirical Evaluation of LLMs as Substitutes for Human Input in Human-Centered Computing

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Human-Centered Computing (HCC) research increasingly uses large language models to stand in for human input: to code qualitative data, annotate crowdsourced corpora, simulate study participants, and generate personas. This raises an epistemological question for a field built on interpretation, reflexivity, and situated knowledge: when can an LLM substitute for a human, and when must a human remain in the loop? We answer with a review paired with a new experiment. **Study 1** is a PRISMA systematic review of 55 HCC, HCI, and qualitative-methods studies (all references independently web-verified), coding each on method, the LLM’s intended role (augment / replace / simulate), the reported reliability against a human baseline, and the epistemic stance. The field overwhelmingly positions LLMs to *augment* rather than replace humans (36 augment vs. 14 replace vs. 5 simulate), under a stance dominated by *tension*, with reliability ordered augment > replace > simulate. **Study 2** tests the field’s central boundary empirically: we run a six-model open panel (9B-754B) as virtual annotators on 4 subjective social-computing datasets that carry human label distributions (emotion, offensiveness, hate, irony; 12,000 judgments), benchmarked against the human-human agreement ceiling with chance-corrected agreement, a calibrated contamination screen, and an item-level mixed-effects analysis. The augment-not-replace consensus is empirically warranted: open judges fall short of the human ceiling on every subjective construct, and, tellingly, the judge-vs-ceiling gap is *largest where humans most agree* (−0.23 on low-disagreement items), so the shortfall is a genuine substitution failure, not a label-noise artifact on ambiguous cases. Residual variance is item-level ($SD_{\text{logit}} = 2.326$ vs model 0.36), not model scale or contamination, which we confirm on a recent post-cutoff corpus. We translate this into an evidence-based decision framework for when HCC researchers should automate, augment with a human in the loop, or keep humans.

Additional Key Words and Phrases: human-centered computing, LLM substitution, qualitative coding, crowdsourced annotation, participant simulation, human-AI collaboration, annotator disagreement

1 INTRODUCTION

In barely two years, large language models have moved from objects of HCI study to *instruments* of HCI method. Researchers now use them to code interview transcripts [14, 35], run thematic analyses [9, 25], annotate social-media corpora [15, 32], simulate survey respondents [1], and generate personas [27]. The appeal is obvious: human judgment is the costly bottleneck of human-centered research. The danger is equally clear. HCC and qualitative methods are not mere labeling pipelines; they carry epistemological commitments, interpretation, reflexivity, positionality, and situated knowledge, that an LLM cannot obviously honor [10, 29], and substituting a model for a human participant can flatten the very identity groups HCI seeks to center [8, 34].

The field therefore faces a question with no consolidated answer: **for which HCC tasks is an LLM a safe substitute for human input, for which is it a useful assistant, and for which is it unsafe?** We answer with two complementary studies. First, a systematic review of how HCC *actually* uses LLMs and how reliable that use is. Second, because a review alone cannot settle a contested empirical boundary (and because the field rightly distrusts AI-generated literature surveys without new evidence), we run a controlled experiment on the field’s central case: subjective social-computing annotation, the kind of labeling HCC and CSCW depend on for studies of affect, toxicity, and online behavior. Throughout, we measure LLMs against the *human-human agreement ceiling* computed from each dataset’s own annotators, because for subjective constructs there is no single ground truth [2, 26], and we use *open, locally-servable* models, the realistic choice for most HCC labs.

Contributions. (1) **Study 1:** the first systematic review (PRISMA, 55 studies, all references web-verified) of LLM substitution specifically for HCC/HCI/qualitative methods, coding method $\times$ role (augment/replace/simulate) $\times$ epistemic stance, which surfaces an augment-not-replace consensus and its tensions. (2) **Study 2:** a new empirical evaluation of a six-model open panel on 4 subjective social-computing datasets with human label distributions (12,000 judgments), showing the consensus is empirically warranted and that item-level human disagreement, not model scale, governs reliability. (3) An evidence-based decision framework for HCC practice. All code, corpora, and harnesses are released.

2 STUDY 1: LLM SUBSTITUTION IN HCC, HCI, AND QUALITATIVE METHODS

2.1 Motivation and research questions

Cross-domain evidence (our companion paper) shows LLM-judge reliability is broadly domain-general; HCC needs a method-specific account. Human-centered research is built on methods that historically *require* human input: qualitative coding, thematic and content analysis, interviews and open-ended survey analysis, crowdsourced annotation, participant studies, persona development, usability/heuristic evaluation, and design critique. Study 1 asks: **(RQ1.1)** which of these methods have been attempted with LLMs, and how reliably do LLM outputs agree with human input? **(RQ1.2)** How does the field position the LLM, to *augment*, *replace*, or *simulate* the human? **(RQ1.3)** What epistemic stance (enabling, tension, critical) does the literature take, and what does that imply for HCC practice?

2.2 Method

Following the shared protocol, we ran a review-specific identification arm: 25 HCC/HCI/qualitative concept-block queries against OpenAlex (857 unique works), triangulated with the web arm and a venue-scoped sweep of HCI venues (CHI, CSCW, UIST, IUI, TOCHI). After topicality screening and eligibility assessment we included 55 studies (Figure 1). An independent re-coding of a 30% sample ($n=16$) reproduced the verdict at Krippendorff $\alpha = 0.85$ and the augment/replace/simulate role at $\alpha = 0.80$; the *epistemic stance* code is the least reliable ($\alpha = 0.59$), so we treat stance-based claims as indicative rather than precise (Appendix C). We extracted, per study: the HCC **method**; the LLM **role** (replace / augment / simulate); reported agreement and the human baseline; the reliability **verdict**; the **epistemic stance** (enabling / tension / critical); and the **community** (HCI, CSCW, qualitative-methods, computational social science, NLP, health). The full corpus is Table 6.

2.3 RQ1.1: Which methods, and how reliable?

LLMs have been compared against human input across 15 distinct HCC methods (Figure 2a), most heavily in *thematic analysis*, *synthetic participants*, and *crowdsourced annotation*. Reliability is strongly method-dependent (Figure 2b; Table 1). The *same objectivity/interpretation axis* seen across domains governs HCC: **objective crowd-annotation** is the most reliable method to substitute: ChatGPT matches or beats crowd workers and even trained coders on stance, topic, and frame labeling [15, 32]; **deductive/inductive coding and content/thematic analysis** are reliable *as augmentation*, with human-equivalent agreement on structured codebooks (GPT-4 exceeding substantial κ on 8/9 hermeneutic tasks [12]; Gemini $\kappa = 0.907$ in multi-LLM thematic analysis [19]; 79.7% agreement in health thematic summarization [7]) but only moderate agreement in open interpretive settings ($\kappa \approx 0.40$ - 0.46 [20, 36]); and **participant/persona simulation and usability evaluation** are the least reliable.

Figure S2-1. Study 2 identification (HCC/HCI/qualitative methods)

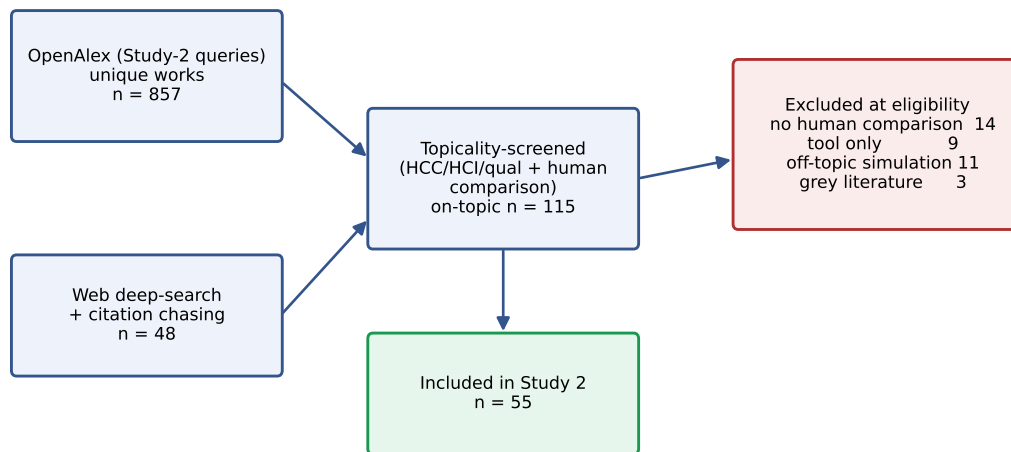


Fig. 1. Study 1 identification flow (HCC/HCI/qualitative methods).

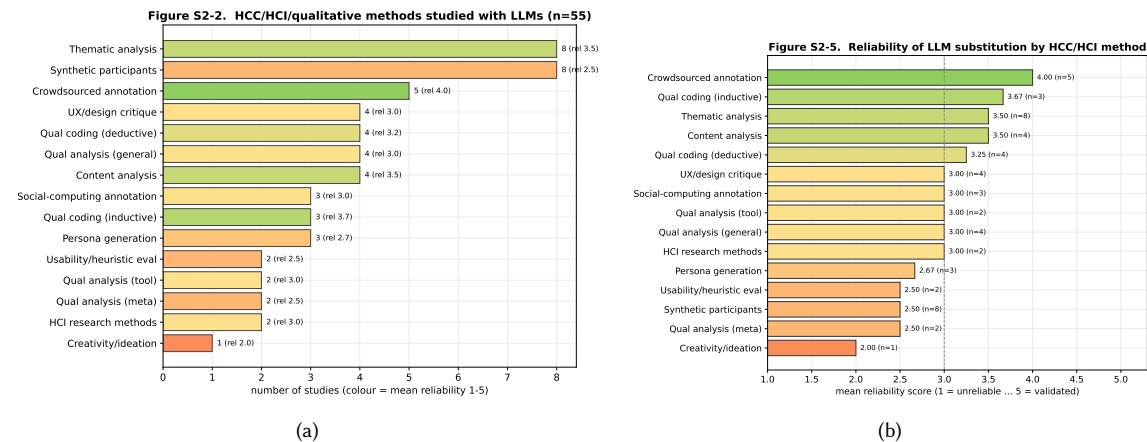


Fig. 2. Study 1. (a) HCC/HCI/qualitative methods studied with LLMs (colour = mean reliability). (b) Mean reliability of LLM substitution by method.

2.4 RQ1.2: Augment, replace, or simulate?

HCC overwhelmingly positions LLMs to *augment* humans, not replace them: 36 of 55 studies frame the LLM as a collaborator/assistant, versus 14 as a replacement and 5 as a human simulator (Figure 3a). This is not merely rhetorical caution; it tracks reliability (Figure 4): augmentation studies skew reliable, whereas *replacement* is validated only for objective annotation and *simulation* of participants is the least reliable role (mean reliability augment 3.17 vs. replace 3.07 vs. simulate 2.8). The replacement mean is buoyed by objective crowd-annotation; remove it and replacement of

Table 1. Study 1 method-level decision table: study count, modal LLM role, mean reliability, and traffic-light signal (GREEN ≥ 3.5 ; AMBER 2.5-3.5; RED < 2.5).

Method	<i>n</i>	Modal role	Mean rel.	Signal
Crowdsourced annotation	5	replace	4.00	GREEN
Qual coding (inductive)	3	augment	3.67	GREEN
Content analysis	4	augment	3.50	GREEN
Thematic analysis	8	augment	3.50	GREEN
Qual coding (deductive)	4	augment	3.25	AMBER
HCI research methods	2	augment	3.00	AMBER
Qual analysis (general)	4	augment	3.00	AMBER
Qual analysis (tool)	2	augment	3.00	AMBER
Social-computing annotation	3	replace	3.00	AMBER
UX/design critique	4	augment	3.00	AMBER
Persona generation	3	augment	2.67	AMBER
Qual analysis (meta)	2	augment	2.50	AMBER
Synthetic participants	8	simulate	2.50	AMBER
Usability/heuristic eval	2	replace	2.50	AMBER
Creativity/ideation	1	augment	2.00	RED

Figure S2-3. How HCC research positions LLMs: augment vs replace, and epistemic stance

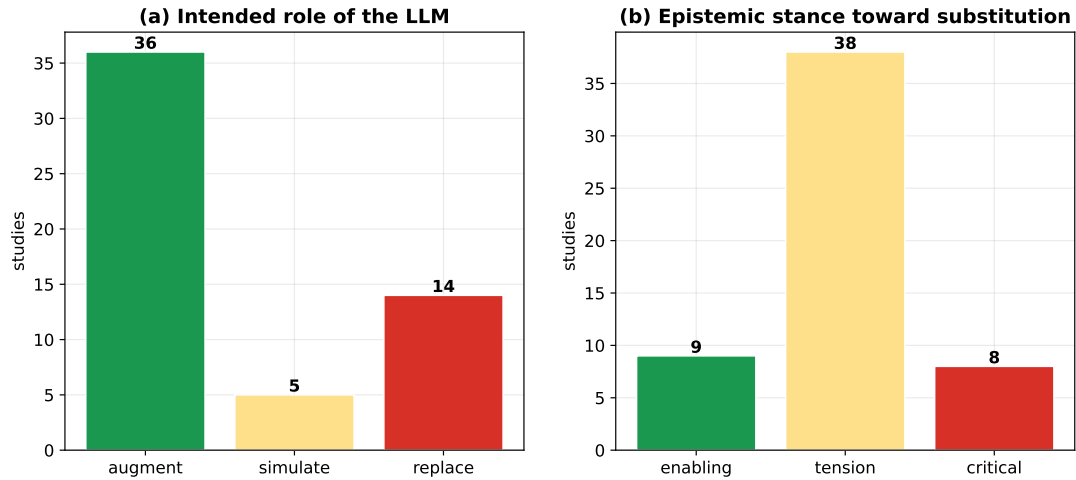


Fig. 3. Study 1. (a) Intended LLM role across the HCC corpus; augmentation dominates. (b) Epistemic stance toward substitution; *tension* dominates.

interpretive human input is firmly amber/red. The canonical HCC workflow that emerges is human-in-the-loop: LLM proposes, human vets and retains interpretive authority [9, 14, 22].

2.5 RQ1.3: Epistemic stance and the limits of substitution

The dominant stance is *tension* (38/55), with 8 explicitly *critical* and only 9 unreservedly *enabling* (Figure 3b). Three threads recur. (i) **Interpretation and reflexivity**. Qualitative scholars warn that LLMs perform surface-level extraction well but cannot supply the situated, reflexive interpretation that gives qualitative findings their validity, and that

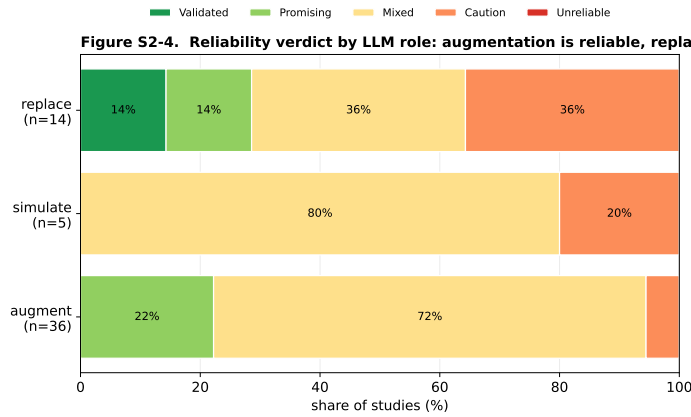


Fig. 4. Study 1. Reliability verdict by LLM role. Augmentation is reliable; replacement is reliable only for objective annotation; participant/usability simulation is the least reliable.

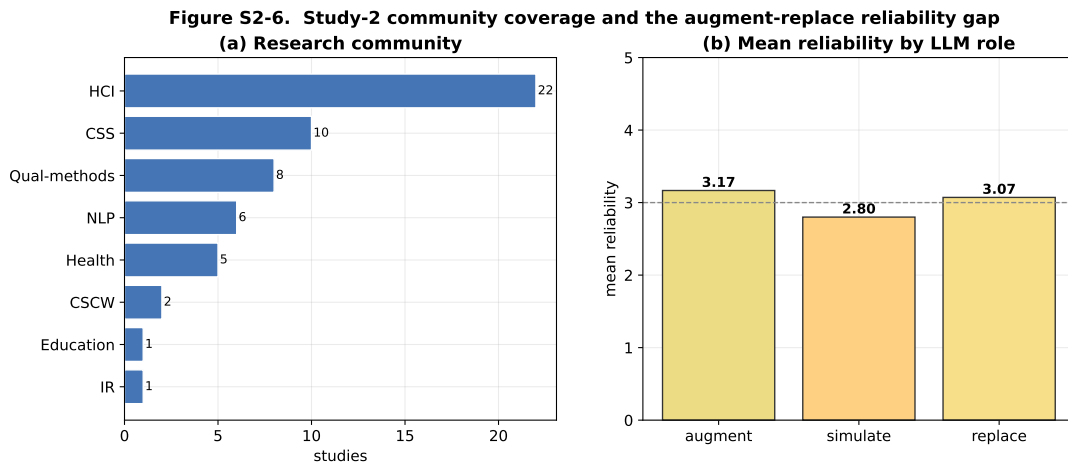


Fig. 5. Study 1. (a) Coverage by research community. (b) Mean reliability by LLM role, augmentation > replacement > simulation.

“conditional trust” in tool output risks hollowing out analytic rigor [10, 23, 24, 29]. (ii) **Participants and identity.** Replacing human participants with LLMs *flattens and misportrays* demographic groups, positionality cannot be simulated [34], and LLM “subgroups” are often caricatures [8]; synthetic survey data distort variances and correlations [5], and an LLM is worth only a bounded number of real respondents [18]. HCI critiques accordingly recast synthetic users as “objects of inquiry,” not stand-ins [17, 28]. (iii) **Expert design judgment.** GPT-4o overlaps only 21% with expert-found usability issues and hallucinates false positives [16], and LLM UI critique trails expert designers [11], though synthetic heuristic evaluation can exceed *novice* evaluators on issue recall [37]. Figure 5 shows coverage skews to HCI and computational social science, with the augment-replace reliability gap visible across communities.

2.6 Study 1 synthesis

For HCC research, the evidence supports a clear, conditional posture. **Validated to substitute:** objective crowd/data annotation (stance, topic, relevance, frames) where a human ceiling is modest and verifiable. **Reliable only as augmentation (human-in-the-loop):** deductive/inductive qualitative coding and thematic/ content analysis: LLMs accelerate breadth and first-pass coding, but humans must own interpretation, theme development, and reflexive accounting. **Unsafe to substitute:** simulating human participants, generating personas as stand-ins for real users, and replacing expert usability/design judgment: these flatten identity, fabricate issues, and forfeit the situated knowledge that defines human-centered inquiry. This posture both confirms the cross-domain objective/subjective prediction *within* HCC and exposes concrete gaps (§3) where HCC datasets with human labels have never been formally tested with an LLM judge.

3 STUDY 2: AN EMPIRICAL TEST ON SUBJECTIVE SOCIAL-COMPUTING ANNOTATION

The review establishes what the field *believes* (augment, do not replace) but not whether that belief is empirically warranted on the tasks HCC depends on. Study 2 supplies the missing evidence on the central case: subjective social-computing annotation. We evaluate a panel of 6 open models (9B-754B, served via an institutional vLLM API at temperature 0) as virtual annotators on 4 datasets that ship human label distributions, the constructs HCC and CSCW routinely label: **emotion** (GoEmotions, an authors-defined 4-way sentiment grouping), binary **offensiveness** (SBIC), 3-way **hate speech** (HateXplain), and binary **irony** (MultiPICO, a 2024 perspectivist corpus). The run comprised 12,000 judgments with zero API errors and a 100% parse rate.

3.1 Method

Each model labels every item under the same guidelines the human annotators received. We benchmark against the *human-human ceiling* (random-annotator-vs-majority accuracy with an annotator-bootstrap 95% CI) and report accuracy, Cohen’s κ , macro-F1, a majority **jury** [33], and a calibrated contamination screen. Because subjective labels are not a single “gold” [2, 26], we also report disagreement-aware metrics: panel accuracy stratified by item human-entropy, and the Jensen-Shannon divergence between the panel and human label distributions. Dataset cards, prompts, and metric definitions are in Appendices A to D.

3.2 The augment-not-replace consensus is empirically warranted

On every subjective construct the open panel falls short of the human ceiling (Table 2). Binary offensiveness is the strongest case (**Promising**; best 0.86) yet still below its 0.91 ceiling; multi-way emotion and 3-way hate are **Mixed** (best 0.61 / 0.66); and even where headline accuracy looks high, it reflects base-rate learning rather than construct competence: on the recent irony corpus the best model reaches 0.82 accuracy but only $\kappa=0.40$. Chance-corrected agreement (Table 3) is modest throughout, exactly the pattern the review’s augment-not-replace consensus predicts.

3.3 The shortfall is largest where humans agree, and is not a label-noise artifact

A natural worry is that judges look worse on high-disagreement items simply because the majority gold is noisier there (a tautology). The judge-vs-ceiling *gap* rules this out. Raw accuracy does fall on high-entropy items (0.77 low-entropy vs 0.54 high-entropy), but the gap below the human ceiling is in fact *largest on the low-entropy items where humans most agree* (-0.23 vs -0.04): on clear-consensus social items, where there is a definite human answer, the panel still

Table 2. Study 2. Open panel vs the human-human ceiling on 4 subjective social-computing datasets. Ceiling = random-annotator-vs-majority accuracy; macro-F1 is the best model’s chance-corrected score.

Dataset	Construct	Human ceiling	Best model (acc)	macro-F1 (best)	Jury (acc)	Modal verdict
GoEmotions	emotion (4-way)	0.78	0.61 (qwen3.6-27b)	0.54	0.61	Mixed
SBIC	offensive (binary)	0.91	0.86 (glm-5.1)	0.85	0.84	Promising
HateXplain	hate (3-way)	0.83	0.66 (gemma-4-31b)	0.64	0.62	Mixed
MultiPICO	irony (binary)	0.82	0.82 (gemma-4-31b)	0.70	0.83	Promising

Table 3. Study 2 disagreement-aware results. Ceiling with annotator-bootstrap 95% CI; best-model κ and macro-F1; panel accuracy on low- vs high-human-disagreement items; Jensen-Shannon panel-vs-human divergence. Judges lose accuracy precisely as human disagreement rises.

Dataset	Ceiling [95% CI]	κ (best)	macro-F1 (best)	$\text{acc}_{\downarrow H}$ low-disagr.	$\text{acc}_{\uparrow H}$ high-disagr.	JS
GoEmotions	0.78 [0.76,0.80]	0.44	0.54	0.60	0.45	0.31
SBIC	0.91 [0.89,0.92]	0.71	0.85	0.86	0.61	0.10
HateXplain	0.83 [0.82,0.85]	0.49	0.64	0.66	0.48	0.27
MultiPICO	0.82 [0.81,0.83]	0.40	0.70	0.80	0.63	0.09

Table 4. Study 2. Judge-vs-ceiling gap (panel accuracy minus human ceiling) by item human-disagreement, per dataset, with annotator/item-bootstrap 95% CIs. The gap is largest (most negative) where humans most agree.

Dataset	gap, low-disagreement (judge – ceiling) [95% CI]	gap, high-disagreement (judge – ceiling) [95% CI]
GoEmotions	-0.24 [-0.28, -0.20]	-0.11 [-0.18, -0.03]
SBIC	-0.14 [-0.16, -0.12]	-0.06 [-0.12, +0.00]
HateXplain	-0.34 [-0.38, -0.30]	-0.19 [-0.23, -0.14]
MultiPICO	-0.12 [-0.14, -0.10]	-0.03 [-0.07, +0.01]

misses it by a wide margin. This is a genuine substitution failure, not noise on ambiguous cases. The residual variance is item-level: in an item-level GLMM the random-intercept SD across *items* ($\text{SD}_{\text{logit}} = 2.326$) far exceeds that across *models* ($\text{SD}_{\text{logit}} = 0.36$, on the same logit scale), and standardized model scale contributes only modestly ($\text{OR}_{\text{SD}} = 1.21$); with four datasets we do not fit a dataset-level model. Contamination is unlikely to explain the shortfall: the recent (2024) MultiPICO corpus, which the low-recall judge has not memorized, is matched only at its low ceiling, so the limitation tracks human judgment, not pretraining exposure (the verbatim screen is conservative, Appendix D).

The gap pattern is consistent and statistically reliable per dataset (Table 4): on all four datasets the judge-vs-ceiling gap is significantly more negative on low-disagreement items than on high-disagreement items (non-overlapping 95% CIs for GoEmotions and HateXplain), confirming that the shortfall is concentrated where humans agree, not where the gold is noisy.

What this means for HCC. The model that simply predicts the majority looks accurate but erases the minority readings that perspectivist annotation is designed to preserve, which is precisely the flattening HCC warns against [13, 34]. Empirically, then, the field’s instinct is right: for subjective social annotation an LLM is a useful first-pass assistant but not a replacement for the human annotators whose disagreement is the signal.

4 DISCUSSION

4.1 Review and experiment agree

The review and the experiment tell one story. The field positions LLMs to *augment* rather than replace humans, with reliability ordered $\text{augment} > \text{replace} > \text{simulate}$; the experiment shows why that ordering is correct for the field’s central labeling case. On subjective social-computing constructs, open judges do not reach the human-human ceiling, and their errors are not random: they concentrate on the items where humans themselves disagree. The binding constraint is therefore not model capability but the contested human reference, which is exactly the property that makes these constructs “human-centered” in the first place.

4.2 Connections to HCI and measurement scholarship

Three HCI conversations are sharpened by this evidence. **(1) There is no single ground truth.** CSCW and HCI have long argued that inter-rater reliability is a negotiated practice, not a natural fact [21], and NLP has reframed annotator disagreement as signal [2, 26]; benchmarking against the human ceiling makes the judges’ “failures” legible as properties of the construct, not fixable model deficits. **(2) Interpretation is not labeling.** Reflexive thematic analysis treats coding as situated interpretation [6], and qualitative scholars warn that “conditional trust” in LLM output hollows out rigor [10, 29]; the augment-not-replace consensus and the empirical shortfall are two views of the same boundary. **(3) Participants are not simulable.** The “silicon sample” program [1] meets a growing critique that simulated participants flatten identity [5, 8, 28, 34]; our finding that judges default to majority readings and lose the minority signal is that flattening at the level of a single subjective label, a representational harm a single accuracy number conceals [4, 13].

4.3 A critical assessment for the field

We name two risks the community should track, marking what our data support and what extends prior critique. **False closure (data-grounded).** Reporting one accuracy against a majority gold imposes consensus where the data shows disagreement; our irony case (0.82 accuracy at $\kappa=0.40$) shows how a headline number masks base-rate learning. **Monoculture and deskilling (conjecture).** Cheap labeling invites substituting LLM judgment for the human annotation and reflexive interpretation that train researchers; combined with the documented convergence of judges toward crowd rather than expert consensus [3] and the way technical framings abstract away the social context of measurement [30, 31], this could erode the field’s interpretive capacity. We offer it as a hypothesis, not a finding.

4.4 A decision framework for HCC

- **Automate (with light audit):** objective annotation of user-generated content (stance, topic, frames) where the human ceiling is high and verifiable [15, 32].
- **Augment, human-in-the-loop:** subjective coding and thematic/content analysis: use the LLM for first-pass coding, breadth, and disagreement surfacing, but keep human interpretive authority, calibrate on a labeled subset, and report chance-corrected agreement against human coders [9, 14].
- **Keep humans (do not substitute):** simulating participants or personas, and replacing expert usability/design judgment, which flatten identity and forfeit situated knowledge [16, 28, 34].

Practically, any HCC study reporting LLM-assisted analysis should state the LLM’s *role*, benchmark against a *human-human* baseline at the right granularity, report chance-corrected and disagreement-aware metrics with uncertainty, evaluate on post-cutoff data where possible, and treat minority annotator readings as signal rather than noise.

5 LIMITATIONS

The *human* validation here is inherited from prior work, not added by us: Study 2’s ground truth is each dataset’s own human annotators (3 to 8 per item), and the review codes each study against the human baselines those studies report. The review verdict is our interpretation (only a minority of studies report an extractable numeric agreement value); we report an independent re-coding as a reproducibility check (verdict Krippendorff $\alpha = 0.85$, role $\alpha = 0.80$, the more interpretive stance $\alpha = 0.59$; Appendix C), which is LLM-assisted and therefore measures rubric reproducibility; we treat stance claims as indicative. All references were web-verified. By design we evaluate *open* models (the reproducible choice for most HCC labs) at temperature 0 across 4 subjective datasets; interpretive qualitative coding, participant simulation, and design judgment are characterized by the review rather than tested empirically here. Findings reflect models available through mid-2026.

6 ETHICS AND REPRODUCIBILITY

Ethics. The review analyzes published research, not human subjects, and Study 2 reuses already-public, de-identified datasets under their licenses; this work was therefore not human-subjects research requiring IRB review, and we sought an exemption determination consistent with our institution’s policy. Two datasets (HateXplain, SBIC) contain hateful or identity-attacking text, which we process only to measure moderation reliability and never reproduce; to avoid redistributing harmful or personally-identifiable content we release only item-level labels and aggregate statistics, never the raw texts. A deeper ethical point bears on our own method: by treating the *majority* human label as “ground truth” for the ceiling, we risk encoding majority-group bias and erasing minority readings, exactly the harm we warn against. We mitigate this by (i) reporting the full disagreement (soft labels, entropy) rather than only the majority, (ii) checking, where annotator demographics exist (MultiPICO), that the panel does not agree disproportionately with one subgroup (gap = 0.01; Appendix D), and (iii) recommending disagreement-aware evaluation in our guidance; we note the other corpora lack annotator demographics, so this check is partial. The contamination probe is reported precisely so “validated” verdicts are not silently inflated by pretraining exposure. The work is dual-use only in the benign sense that better-calibrated LLM judges reduce the risk of unaccountable automated moderation; our central finding (that LLM judges should not replace humans on subjective social annotation) is itself a guard against premature deployment.

Reproducibility. Every number in this paper regenerates from released code: the OpenAlex, Semantic Scholar, and web search logs; the coded corpus; the figure and table generators; the Study-2 judging/scoring/contamination/mixed-effects harness, all keyed to documented model identifiers served through an institutional vLLM API at temperature 0 with idempotent, resumable checkpoints. The auto-generated statistics macros mean the manuscript cannot drift from the data. Repository: a public repository (URL omitted for anonymous review).

7 CONCLUSION

Human-Centered Computing is right to position LLMs as assistants rather than replacements for human input. Across 55 reviewed studies and a 12,000-judgment experiment on subjective social-computing annotation, the evidence converges: open LLM judges do not reach the human-human agreement ceiling on contested constructs, and their errors concentrate on exactly the items where humans disagree, which is the signal HCC cares about. The field should treat an LLM judge as a measured instrument, reported against a human-human baseline with chance-corrected, disagreement-aware metrics, used to augment and accelerate human interpretation rather than to stand in for the humans whose disagreement constitutes the phenomenon under study.

REFERENCES

- [1] Argyle et al. 2023. Out of One, Many: Using Language Models to Simulate Human Samples. *arXiv preprint arXiv:2209.06899* (2023). arXiv:2209.06899
- [2] Lora Aroyo and Chris Welty. 2015. Truth Is a Lie: Crowd Truth and the Seven Myths of Human Annotation. In *AI Magazine*, Vol. 36. 15–24.
- [3] Bavaresco et al. 2024. LLMs instead of Human Judges? A Large Scale Empirical Study across 20 NLP Evaluation Tasks. *arXiv preprint arXiv:2406.18403* (2024). arXiv:2406.18403
- [4] Abeba Birhane, Pratyusha Kalluri, Dallas Card, William Agnew, Ravit Dotan, and Michelle Bao. 2022. The Values Encoded in Machine Learning Research. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAccT)*. 173–184.
- [5] Bisbee et al. 2023. Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *SocArXiv* (2023). <https://doi.org/10.31235/osf.io/5ecfa>
- [6] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101.
- [7] Castellanos et al. 2025. Large Language Models for Thematic Summarization in Qualitative Health Care Research: Comparative Analysis of Model and Human Performance. *JMIR AI* (2025). <https://doi.org/10.2196/64447>
- [8] Cheng et al. 2023. CoMPoS: Characterizing and Evaluating Caricature in LLM Simulations. In *EMNLP*. <https://doi.org/10.18653/v1/2023.emnlp-main.669>
- [9] Dai et al. 2023. LLM-in-the-loop: Leveraging Large Language Model for Thematic Analysis. In *EMNLP-F*. <https://doi.org/10.18653/v1/2023.findings-emnlp.669>
- [10] Davison et al. 2024. The Ethics of Using Generative AI for Qualitative Data Analysis. *Information Systems J.* (2024). <https://doi.org/10.1111/isj.12504>
- [11] Duan et al. 2024. UICrit: Enhancing Automated Design Evaluation with a UI Critique Dataset. *arXiv preprint arXiv:2407.08850* (2024). arXiv:2407.08850
- [12] Dunivin. 2024. Scalable Qualitative Coding with LLMs: Chain-of-Thought Reasoning Matches Human Performance in Some Hermeneutic Tasks. *arXiv preprint arXiv:2401.15170* (2024). arXiv:2401.15170
- [13] Miranda Fricker. 2007. *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford University Press.
- [14] Gao et al. 2024. CollabCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models. *arXiv preprint arXiv:2304.07366* (2024). arXiv:2304.07366
- [15] Gilardi et al. 2023. ChatGPT Outperforms Crowd Workers for Text-Annotation Tasks. *arXiv preprint arXiv:2303.15056* (2023). arXiv:2303.15056
- [16] Guerino et al. 2025. Can GPT-4o Evaluate Usability Like Human Experts? A Comparative Study on Issue Identification in Heuristic Evaluation. *arXiv preprint arXiv:2506.16345* (2025). arXiv:2506.16345
- [17] Haxvig et al. 2025. I’ve Never Seen a Glass Ceiling Better Represented: Bias and Gendering in LLM-Generated Synthetic Personas from a Participatory Design Perspective. *Int. J. Hum.-Comput. Stud.* (2025). <https://doi.org/10.1016/j.jihcs.2025.103651>
- [18] Huang et al. 2025. How Many Human Survey Respondents is a Large Language Model Worth? An Uncertainty Quantification Perspective. *arXiv preprint arXiv:2502.17773* (2025). arXiv:2502.17773
- [19] Jain et al. 2025. Multi-LLM Thematic Analysis with Dual Reliability Metrics: Combining Cohen’s Kappa and Semantic Similarity for Qualitative Research Validation. *arXiv preprint arXiv:2512.20352* (2025). arXiv:2512.20352
- [20] Li et al. 2024. Comparing GPT-4 and Human Researchers in Health Care Data Analysis: Qualitative Description Study. *J. Med. Internet Res.* (2024). <https://doi.org/10.2196/56500>
- [21] Nora McDonald, Sarita Schoenebeck, and Andrea Forte. 2019. Reliability and Inter-rater Reliability in Qualitative Research: Norms and Guidelines for CSCW and HCI Practice. In *Proceedings of the ACM on Human-Computer Interaction (CSCW)*, Vol. 3. ACM, 1–23.
- [22] Meng et al. 2024. Exploring the Human-LLM Synergy in Advancing Theory-driven Qualitative Analysis (CHALET). *arXiv preprint arXiv:2405.05758* (2024). arXiv:2405.05758
- [23] Ngo et al. 2026. Qualitative Coding Analysis through Open-Source Large Language Models: A User Study and Design Recommendations. *arXiv preprint arXiv:2602.18352* (2026). arXiv:2602.18352
- [24] Nguyen-Trung. 2025. ChatGPT in Thematic Analysis: Can AI Become a Research Assistant in Qualitative Research? *Quality & Quantity* (2025). <https://doi.org/10.1007/s11135-025-02165-z>
- [25] De Paoli. 2023. Performing an Inductive Thematic Analysis of Semi-Structured Interviews with a Large Language Model. *Soc. Sci. Comput. Rev.* (2023). <https://doi.org/10.1177/08944393231220483>
- [26] Barbara Plank. 2022. The “Problem” of Human Label Variation: On Ground Truth in Data, Modeling and Evaluation. *Proceedings of EMNLP* (2022), 10671–10682.
- [27] Salminen et al. 2024. Deus Ex Machina and Personas from Large Language Models: Investigating the Composition of AI-Generated Persona Descriptions. In *CHI*. <https://doi.org/10.1145/3613904.3642036>
- [28] Salminen et al. 2025. The Use of Large Language Models in HCI: A Critical Analysis of Synthetic Users. *Augmented Humans* (2025). <https://doi.org/10.1145/3745900.3746108>
- [29] Schroeder et al. 2025. Large Language Models in Qualitative Research: Uses, Tensions, and Intentions. In *CHI*. <https://doi.org/10.1145/3706598.3713120>
- [30] Andrew D. Selbst, Danah Boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. Fairness and Abstraction in Sociotechnical Systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAccT)*. 59–68.
- [31] Lucy Suchman. 2007. *Human-Machine Reconfigurations: Plans and Situated Actions*. Cambridge University Press.

- [32] Tornberg. 2023. ChatGPT-4 Outperforms Experts and Crowd Workers in Annotating Political Twitter Messages with Zero-Shot Learning. *arXiv preprint arXiv:2304.06588* (2023). arXiv:2304.06588
- [33] Verga et al. 2024. Replacing Judges with Juries: Evaluating LLM Generations with a Panel of Diverse Models. *arXiv preprint arXiv:2404.18796* (2024). arXiv:2404.18796
- [34] Wang et al. 2024. Large Language Models that Replace Human Participants can Harmfully Misportray and Flatten Identity Groups. *arXiv preprint arXiv:2402.01908* (2024). arXiv:2402.01908
- [35] Xiao et al. 2023. Supporting Qualitative Analysis with Large Language Models: Combining Codebook with GPT-3 for Deductive Coding. *arXiv preprint arXiv:2304.10548* (2023). arXiv:2304.10548
- [36] Xiao et al. 2025. Qualitative Coding with GPT-4: Where it Works Better. *J. Learning Analytics* (2025).
- [37] Zhong et al. 2025. Synthetic Heuristic Evaluation: A Comparison between AI- and Human-Powered Usability Evaluation. *arXiv preprint arXiv:2507.02306* (2025). arXiv:2507.02306

A REVIEW PROTOCOL AND RELIABILITY RUBRIC

Identification. The review follows PRISMA 2020. The primary bibliographic source is the **OpenAlex** API; we cross-check with the **Semantic Scholar** Graph API and a structured **web “deep-research”** arm. The review used a 25-query HCC/HCI/qualitative concept-block protocol (the same query strings were issued against every source so the arms are comparable). Concept blocks combine an LLM term (“LLM”, “large language model”, “GPT-4”, “ChatGPT”, “Gemini”), an evaluation/agreement term (“LLM-as-a-judge”, “automatic evaluation”, “annotation”, “inter-annotator agreement”, “human evaluation”), and an HCC method term (“qualitative coding”, “thematic analysis”, “content analysis”, “crowdsourcing”, “participant simulation”, “persona”, “usability evaluation”, “design critique”). Records were deduplicated by DOI and normalized title.

Screening and eligibility. *Inclusion:* peer-reviewed or preprint studies that empirically compare LLM judgments to human judgments on the same items and report (or allow computing) an agreement statistic. *Exclusion:* non-empirical position papers, studies with no human baseline, studies that use LLMs only to generate (not judge) content, and off-topic retrievals. Title/abstract screening was followed by full-text eligibility assessment; per-stage counts appear in the PRISMA flow figure (Figure 1).

Reliability verdict (five-level rubric). Each study is coded against the *human-human agreement ceiling* for its task:

- **1 Validated:** LLM-human agreement is at or above the human-human ceiling (substitution is empirically justified).
- **2 Promising:** agreement approaches the ceiling under specific, reported conditions (model, prompt, aggregation); conditional substitution.
- **3 Mixed:** results are task- or setup-dependent and inconsistent across studies.
- **4 Caution:** agreement is consistently and materially below the ceiling.
- **5 Unreliable:** agreement is near chance or dominated by bias (position, verbosity, self-preference).

We report a reliability score = 6 – verdict and synthesize within metric families (e.g. Cohen’s κ , Krippendorff’s α , Spearman ρ , accuracy) rather than pooling incommensurable effect sizes. The full coded corpus (Appendix E) is released for re-coding.

B STUDY 2: DATASETS, JUDGING PROMPTS, AND METRICS

Datasets. Study 2 uses 4 subjective social-computing datasets (Table 5); each is sampled to $n=500$ items with a clear majority label (ties dropped), using the per-item annotator distributions to compute the human ceiling. MultiPICO (2024) is a recent corpus that serves as a post-cutoff condition.

Table 5. Study 2 dataset cards. Ceiling = random-annotator-vs-majority accuracy.

Dataset	Task	Source	Annot./item	Ceiling
GoEmotions	sentiment (4-way, authors' grouping)	Reddit (Google)	~3-5	0.78
SBIC	offensiveness (binary)	Social Bias Frames	~3	0.91
HateXplain	hate speech (3-way)	Twitter/Gab	3	0.83
MultiPICO	irony (binary)	Reddit/Twitter (2024)	3-8	0.82

Judging prompts. Each model labels every item as a virtual annotator, returning a JSON label on the final line (parsed with a surface-form-to-code map):

```

Emotion: Comment: {text} / Overall sentiment: POSITIVE, NEGATIVE, AMBIGUOUS, or NEUTRAL. ->
{"label": ...}

Offensive: Post: {text} / Is this post offensive (including potentially offensive)? -> {"label":
"offensive"|"not offensive"}

Hate: Post: {text} / HATE SPEECH (identity-targeted), OFFENSIVE (not group-targeted), or NORMAL.
-> {"label": ...}

Irony: {post+reply} / Is the REPLY ironic/sarcastic, or not ironic? -> {"label": "ironic"|"not
ironic"}
```

Metrics. Let an item have A human annotations with majority label y^* . **Human ceiling** is the random-annotator-vs-majority accuracy averaged over items, with an annotator-bootstrap 95% CI. **Accuracy**, **Cohen's κ** , and **macro-F1** are against y^* ; the **jury** is the per-item panel majority [33]; **95% CIs** are nonparametric bootstrap (1000 item resamples). A **contamination screen** (one-third-prefix verbatim reproduction) is reported as a conservative check; a clean-control (Appendix D) shows it false-positives on high-recall models, so we rely on the recent MultiPICO post-cutoff condition. The run recorded *zero* API errors and a 100% answer-parse rate across all 12,000 judgments.

Inferential model. We model each judgment's correctness with an item-level Bayesian logistic GLMM with random intercepts for item and model, reporting variance components on a common logit scale (item $SD_{\text{logit}} = 2.326$, model 0.36). With four datasets we do not fit a dataset-level model. We bin items by human entropy and report the judge-vs-ceiling gap per bin to show the shortfall is not a label-noise artifact (the gap is largest where humans most agree, see the main text).

C INTER-CODER RELIABILITY

Because the reliability verdict and the role/stance codes embed judgement, we re-coded a held-out random sample of each corpus with two *independent* coders who saw only the per-study summaries and the rubric (no original codes, no other context). For the review ($n=16$, a 30% sample) the reliability verdict reproduced at Krippendorff $\alpha = 0.85$, the augment/replace/simulate role at $\alpha = 0.80$, and the epistemic stance at $\alpha = 0.59$ (the least reliable code). *Caveat:* the re-coders are LLM-assisted, like the primary pipeline, so these figures measure rubric *reproducibility* under independent application, not human inter-rater agreement; the moderate stance α is why we treat stance-based claims as indicative.

D ROBUSTNESS AND VALIDITY CHECKS

Prompt robustness. We re-ran a robustness subset (three models, $n=120$ per cell) under three semantically equivalent prompt paraphrases per task. *Accuracy* is fairly prompt-stable (mean SD across paraphrases = 0.03), but the discrete

five-level *verdict* is exactly stable in only 10/18 cells: most instability is a one-level flip for cells sitting near a band boundary, and the smaller model is the most sensitive (worst-case acc SD 0.14), the larger models the least. We therefore report verdicts as robust to *within one level* rather than exact, and flag small-model verdicts on subjective tasks as prompt-sensitive. **Contamination clean control.** We ran the verbatim-recall probe on freshly authored, post-cutoff sentences that no model can have seen. The result is a cautionary one: the high-recall model still scores 0.45 mean overlap (MEDIUM) on this guaranteed-clean text, i.e. it confabulates plausible continuations regardless of memorization, while the low-recall model scores LOW. The probe thus has a high false-positive rate for high-recall models and *cannot* be read as a reliable leakage test; the uniform MEDIUM tiers in Table 2 are largely an artifact of generative fluency. We retain the probe only as a conservative, low-recall-model signal and base our substantive claims on the disagreement analysis instead. **Annotator-subgroup check.** On MultiPICO (which carries annotator demographics), the panel’s agreement with the female- vs male-annotator majority differs by only 0.01, i.e. we did not detect a large majority-group bias in this corpus, though demographic metadata is sparse and this check is limited. **Ceiling vs annotator count.** The random-annotator-vs-majority ceiling depends on how many annotators an item has: subsampling ChaosNLI-SNLI (~100 annotators) to $A=3$ raises its estimated ceiling from 0.75 to 0.81 (fewer annotators yield more unanimous majorities). We therefore compute each dataset’s ceiling at its *native* annotator count and caution that ceilings are not directly comparable across datasets with very different A ; the disagreement-aware analyses (entropy, κ) do not depend on this estimator.

E STUDY 2 CORPUS

Table 6. Study 2 corpus (55 studies): HCC/HCI/qualitative methods using LLMs vs human input.

Study	Method	Role	Community	Agree	Verdict
Chew et al. (2023)	Content analysis	augment	CSS	-	Promising
Leas et al. (2023)	Content analysis	augment	Health	-	Mixed
Long et al. (2024)	Content analysis	augment	Education	-	Mixed
Dunivin (2025)	Content analysis	augment	CSS	-	Promising
Anderson et al. (2024)	Creativity/ideation	augment	HCI	-	Caution
Gilardi et al. (2023)	Crowdsourced annotation	replace	CSS	-	Validated
Tornberg (2023)	Crowdsourced annotation	replace	CSS	-	Validated
Kuzman et al. (2023)	Crowdsourced annotation	replace	NLP	-	Mixed
Li et al. (2023)	Crowdsourced annotation	augment	NLP	-	Promising
Ostyakova et al. (2023)	Crowdsourced annotation	replace	NLP	-	Mixed
Tabone & de Winter (2023)	HCI research methods	augment	HCI	-	Mixed
Aubin Le Quere et al. (2024)	HCI research methods	augment	HCI	-	Mixed
Salminen et al. (2024)	Persona generation	augment	HCI	-	Mixed
Jung et al. (2025)	Persona generation	augment	HCI	-	Mixed
Haxvig et al. (2025)	Persona generation	replace	HCI	-	Caution
Zhang et al. (2023)	Qual analysis (general)	augment	Qual-methods	-	Mixed
Wachinger et al. (2024)	Qual analysis (general)	augment	Health	-	Mixed
Li et al. (2024)	Qual analysis (general)	augment	Health	0.40	Mixed
Mehta et al. (2025)	Qual analysis (general)	augment	Qual-methods	-	Mixed
Davison et al. (2024)	Qual analysis (meta)	augment	CSS	-	Caution
Schroeder et al. (2025)	Qual analysis (meta)	augment	HCI	-	Mixed
Zhang et al. (2024)	Qual analysis (tool)	augment	HCI	-	Mixed
Meng et al. (2024)	Qual analysis (tool)	augment	HCI	-	Mixed
Xiao et al. (2023)	Qual coding (deductive)	augment	HCI	-	Promising
Savelka et al. (2023)	Qual coding (deductive)	augment	NLP	-	Mixed

Study	Method	Role	Community	Agree	Verdict
Zhang et al. (2025)	Qual coding (deductive)	augment	Qual-methods	0.46	Mixed
authors (2025)	Qual coding (deductive)	augment	Qual-methods	0.46	Mixed
Gao et al. (2024)	Qual coding (inductive)	augment	HCI	-	Promising
Dunivin (2024)	Qual coding (inductive)	replace	Qual-methods	-	Promising
Ngo et al. (2026)	Qual coding (inductive)	augment	HCI	-	Mixed
Wu et al. (2023)	Social-computing annotation	replace	CSCW	-	Mixed
Zhu et al. (2023)	Social-computing annotation	replace	CSCW	-	Mixed
Rajashekar et al. (2024)	Social-computing annotation	augment	HCI	-	Mixed
Argyle et al. (2023)	Synthetic participants	simulate	CSS	-	Mixed
Bisbee et al. (2023)	Synthetic participants	replace	CSS	-	Caution
Cheng et al. (2023)	Synthetic participants	simulate	NLP	-	Caution
Wang et al. (2024)	Synthetic participants	replace	HCI	-	Caution
Bail (2024)	Synthetic participants	simulate	CSS	-	Mixed
Abbasiantaeb et al. (2024)	Synthetic participants	simulate	IR	-	Mixed
Huang et al. (2025)	Synthetic participants	simulate	CSS	-	Mixed
Salminen et al. (2025)	Synthetic participants	replace	HCI	-	Caution
De Paoli (2023)	Thematic analysis	augment	CSS	-	Mixed
Dai et al. (2023)	Thematic analysis	augment	NLP	-	Promising
Prescott et al. (2024)	Thematic analysis	augment	Health	-	Mixed
Nguyen-Trung (2025)	Thematic analysis	augment	Qual-methods	-	Mixed
Jain et al. (2025)	Thematic analysis	replace	Qual-methods	0.91	Promising
Castellanos et al. (2025)	Thematic analysis	augment	Health	0.80	Promising
Sharma et al. (2025)	Thematic analysis	augment	HCI	-	Promising
Wen et al. (2025)	Thematic analysis	augment	Qual-methods	-	Mixed
Kocaballi (2023)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Duan et al. (2024)	UX/design critique	augment	HCI	-	Mixed
Guerino et al. (2025)	Usability/heuristic eval	replace	HCI	0.21	Caution
Zhong et al. (2025)	Usability/heuristic eval	replace	HCI	0.75	Mixed